

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

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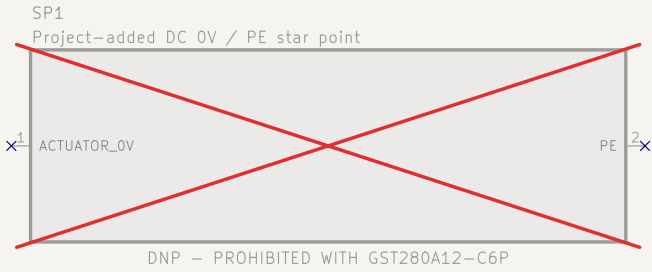
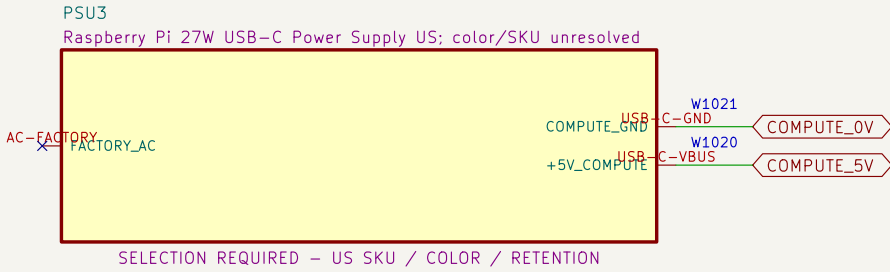
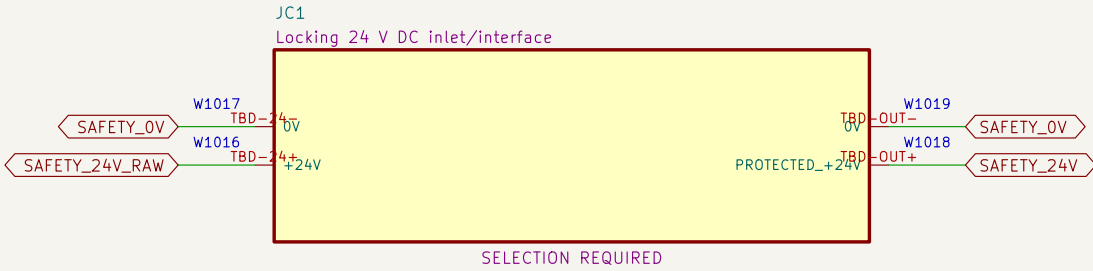
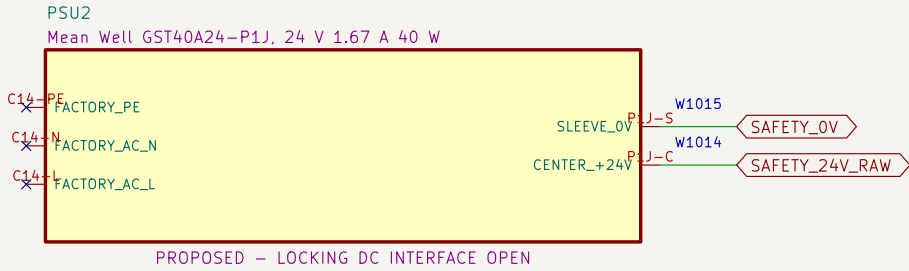
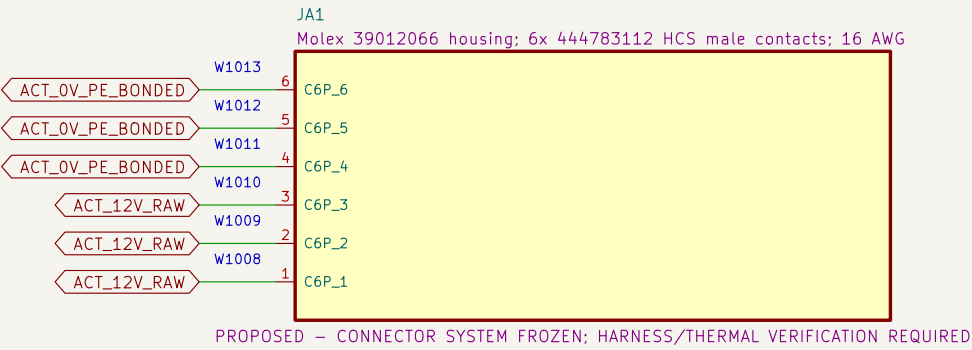
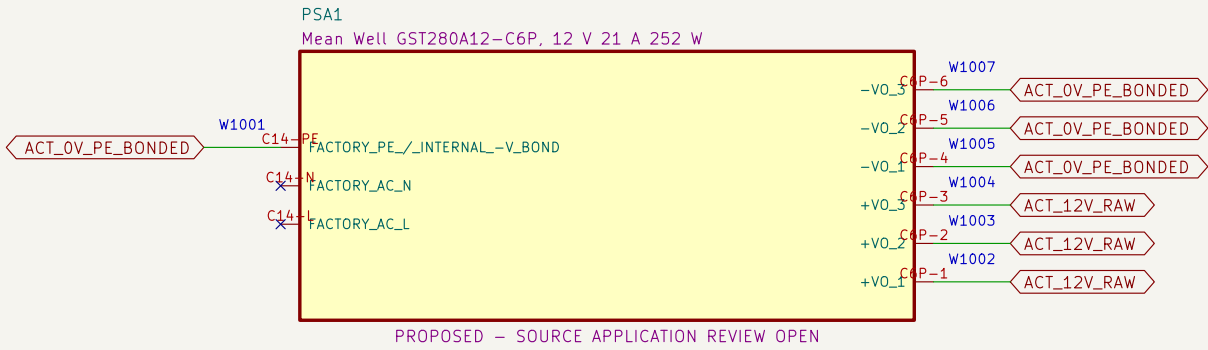
10 U2D2, actuator ports and bonding boundary

File: 10_actuator_interfaces.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button		
Sheet: / File: project–button–v3.kicad_sch		
Title: Project Button HR–V0 Electrical V3 index		
Size: A3	Date: 2026–08–06	Rev: V3–P0.9
KiCad E.D.A. 10.0.5	Id: 1/11	

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.

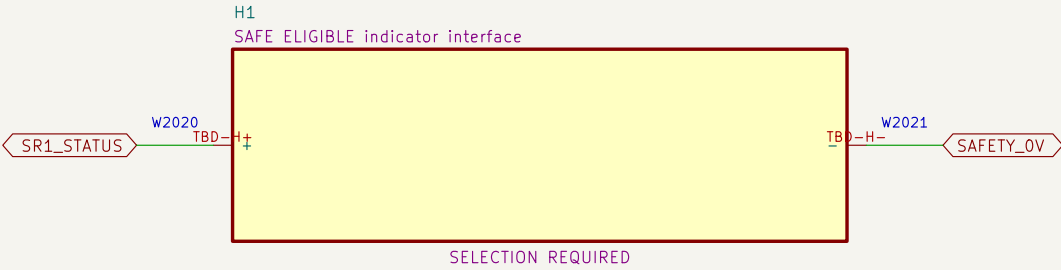
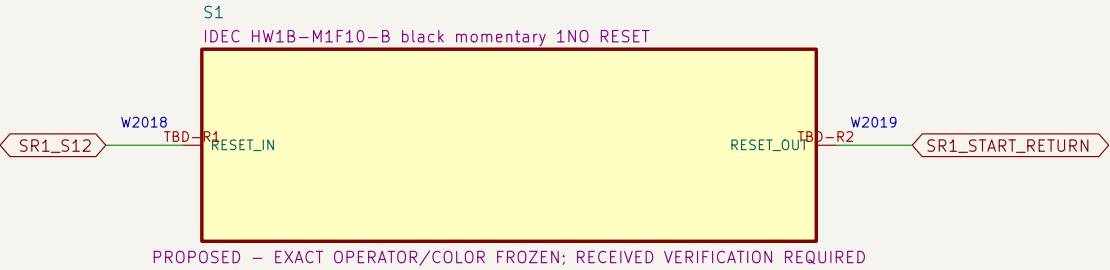
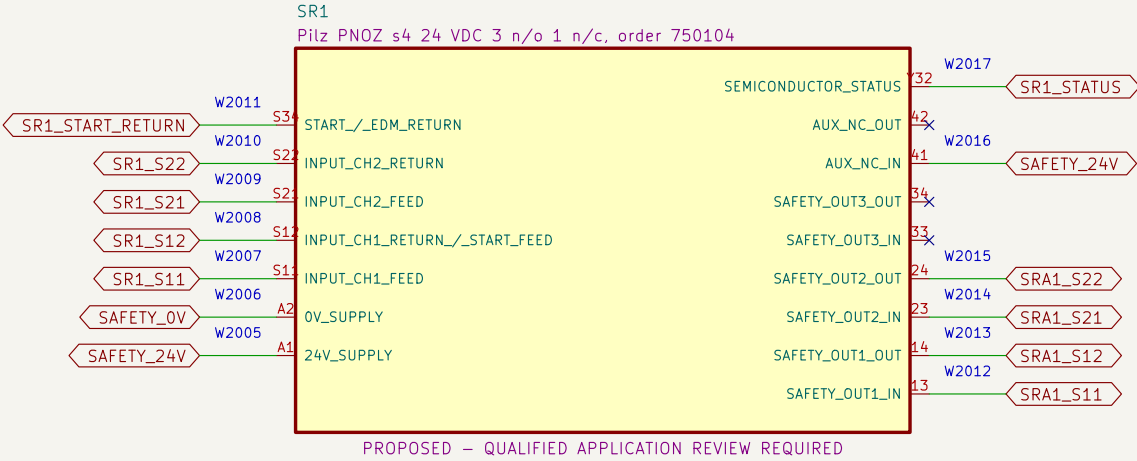
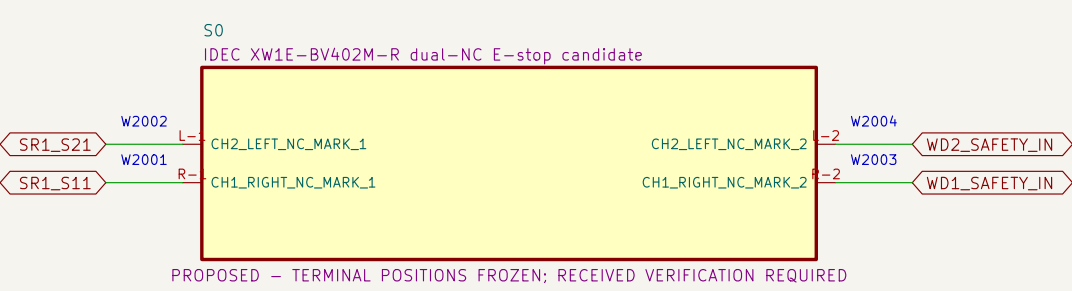


NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



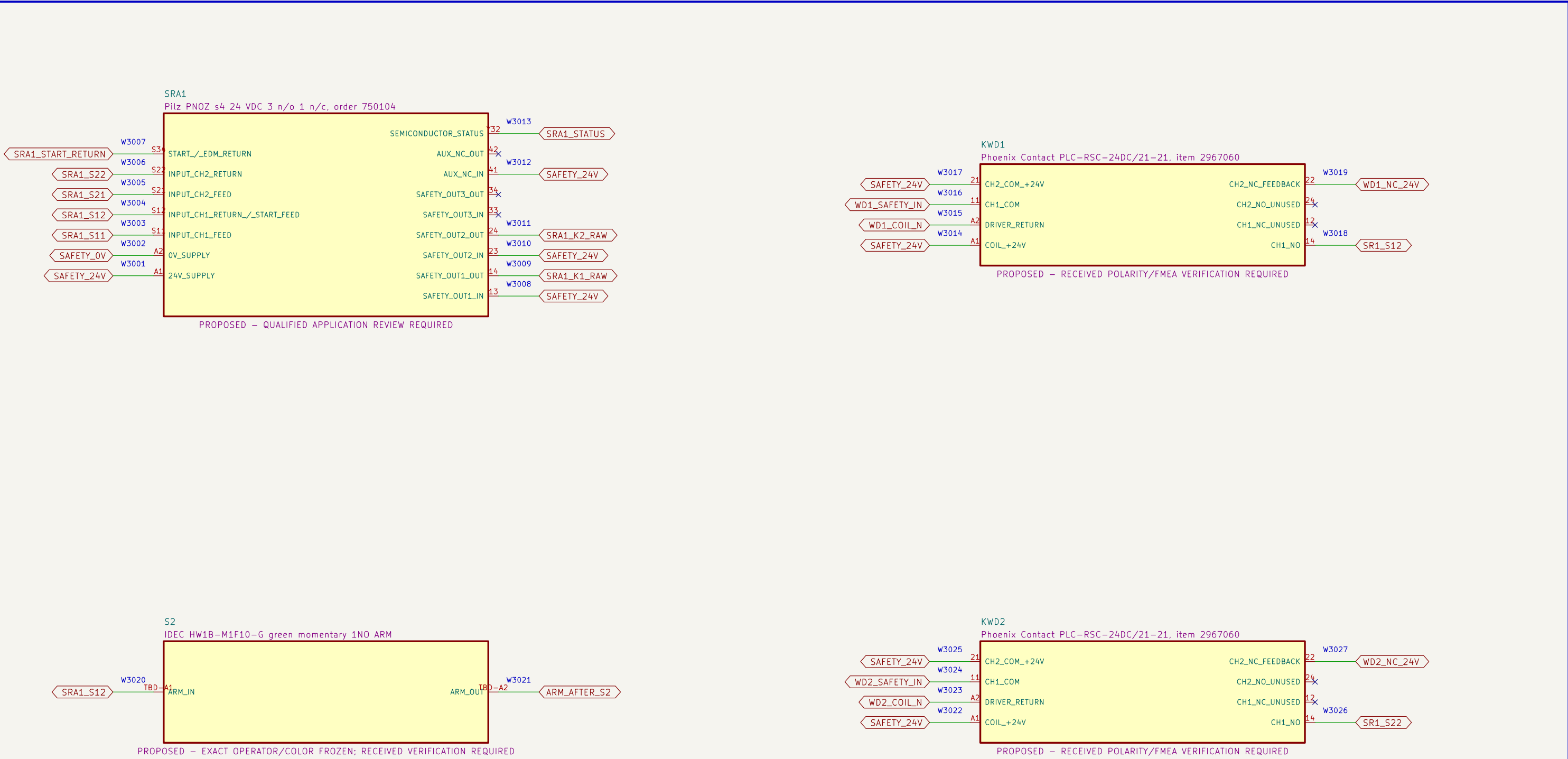
NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /02 Dual-channel E-stop and RESET eligibility/		
File: 02_estop_eligibility.kicad_sch		
Title: PB HR-V0 Electrical V3 – 02		
Size: A3	Date: 2026-08-06	Rev: V3-P0.9
KiCad E.D.A. 10.0.5		Id: 3/11

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



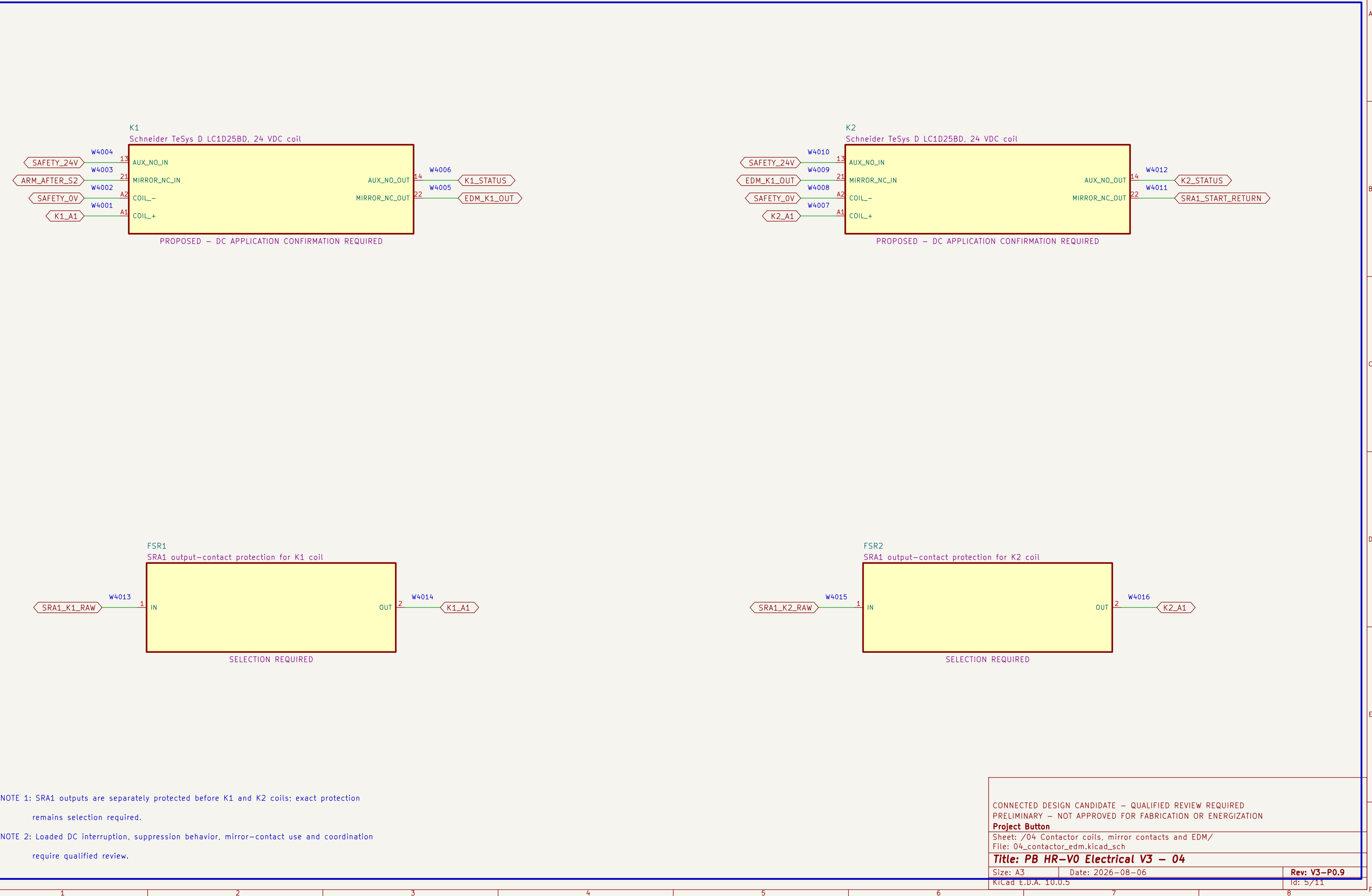
NOTE 1: Required after E–stop/watchdog dropout: cause healthy –> RESET press/release –> SAFE_READY –> distinct ARM press/release.

NOTE 2: KWD contacts are in the SR1 returns; SRA1 receives the two SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /03 Distinct ARM and watchdog eligibility/ File: 03_arm_watchdog_eligibility.kicad_sch		
Title: PB HR–V0 Electrical V3 – 03		
Size: A3	Date: 2026–08–06	Rev: V3–P0.9
KiCad E.D.A. 10.0.5		Id: 4/11

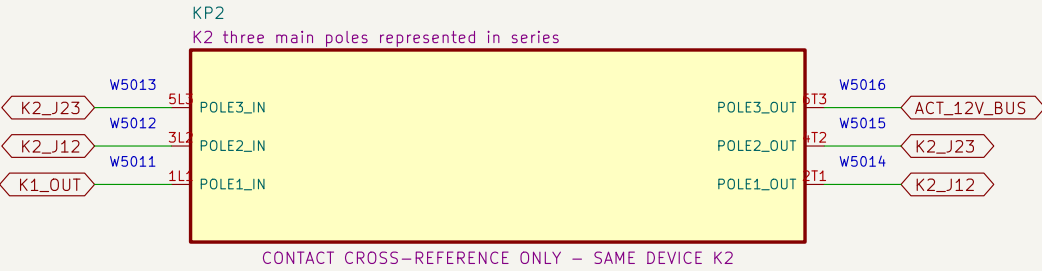
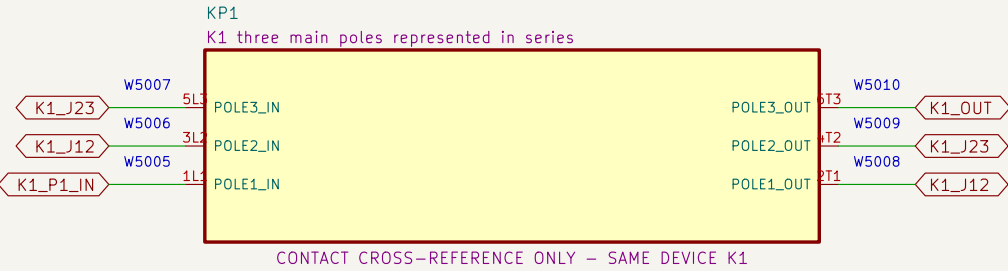
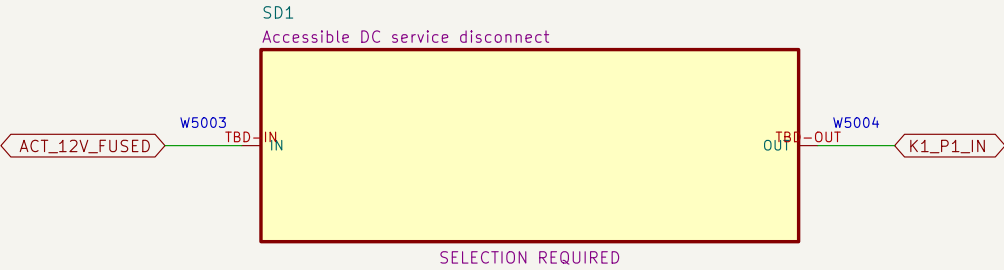
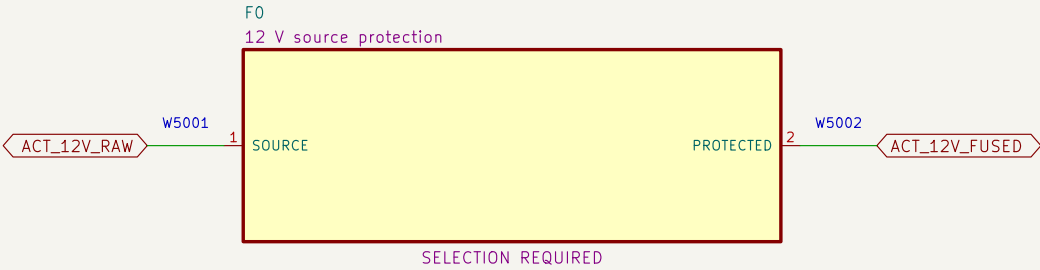
04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



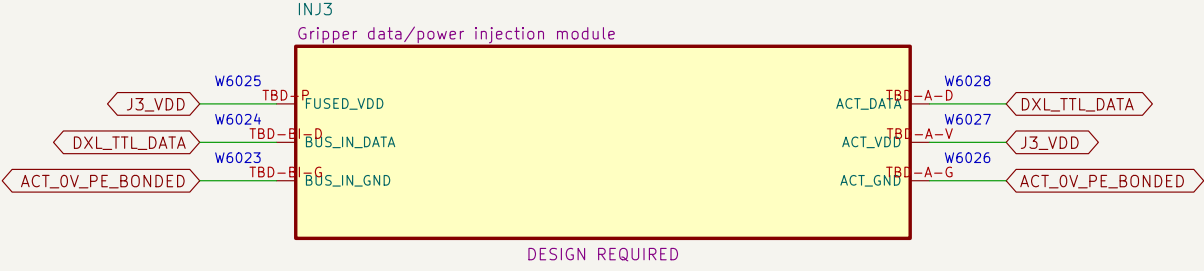
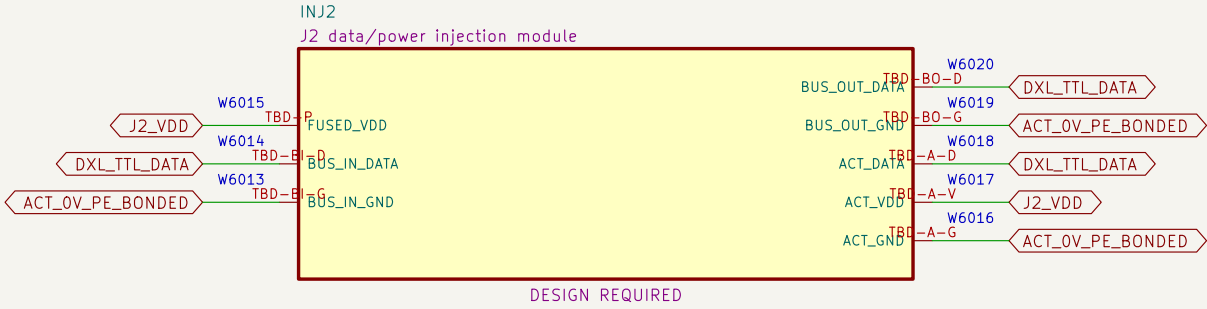
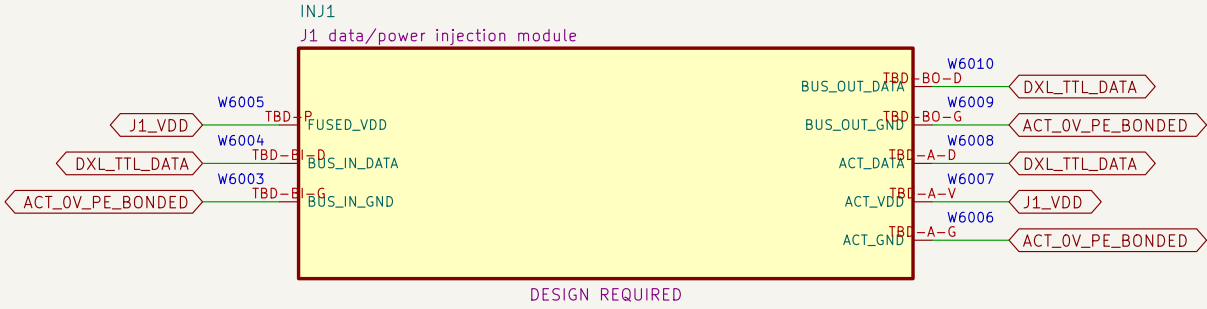
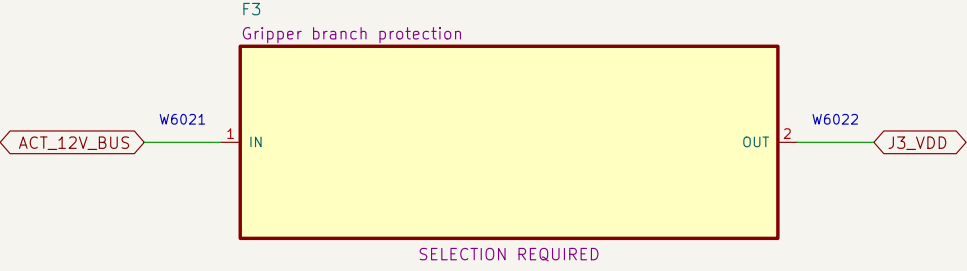
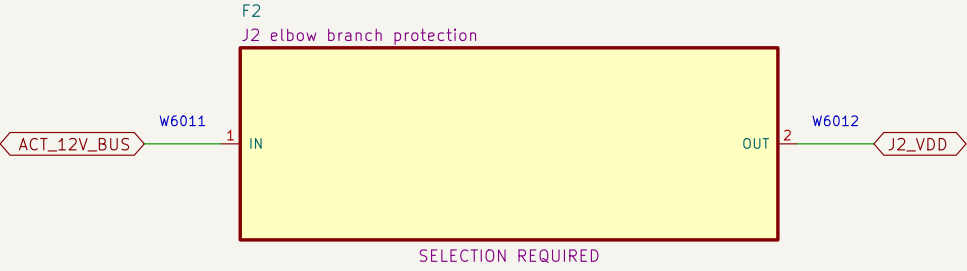
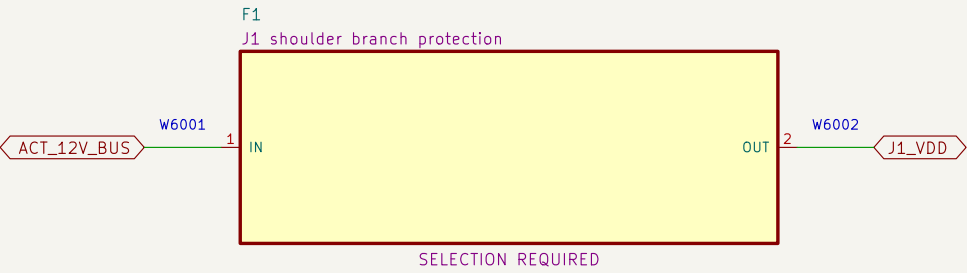
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–V0 Electrical V3 – 05		
Size: A3	Date: 2026–08–06	Rev: V3–P0.9
KiCad E.D.A. 10.0.5		Id: 6/11

06 Protected actuator branches and VDD-isolating injection

Each actuator has a separate protected VDD branch; inter-actuator data cables must omit VDD.



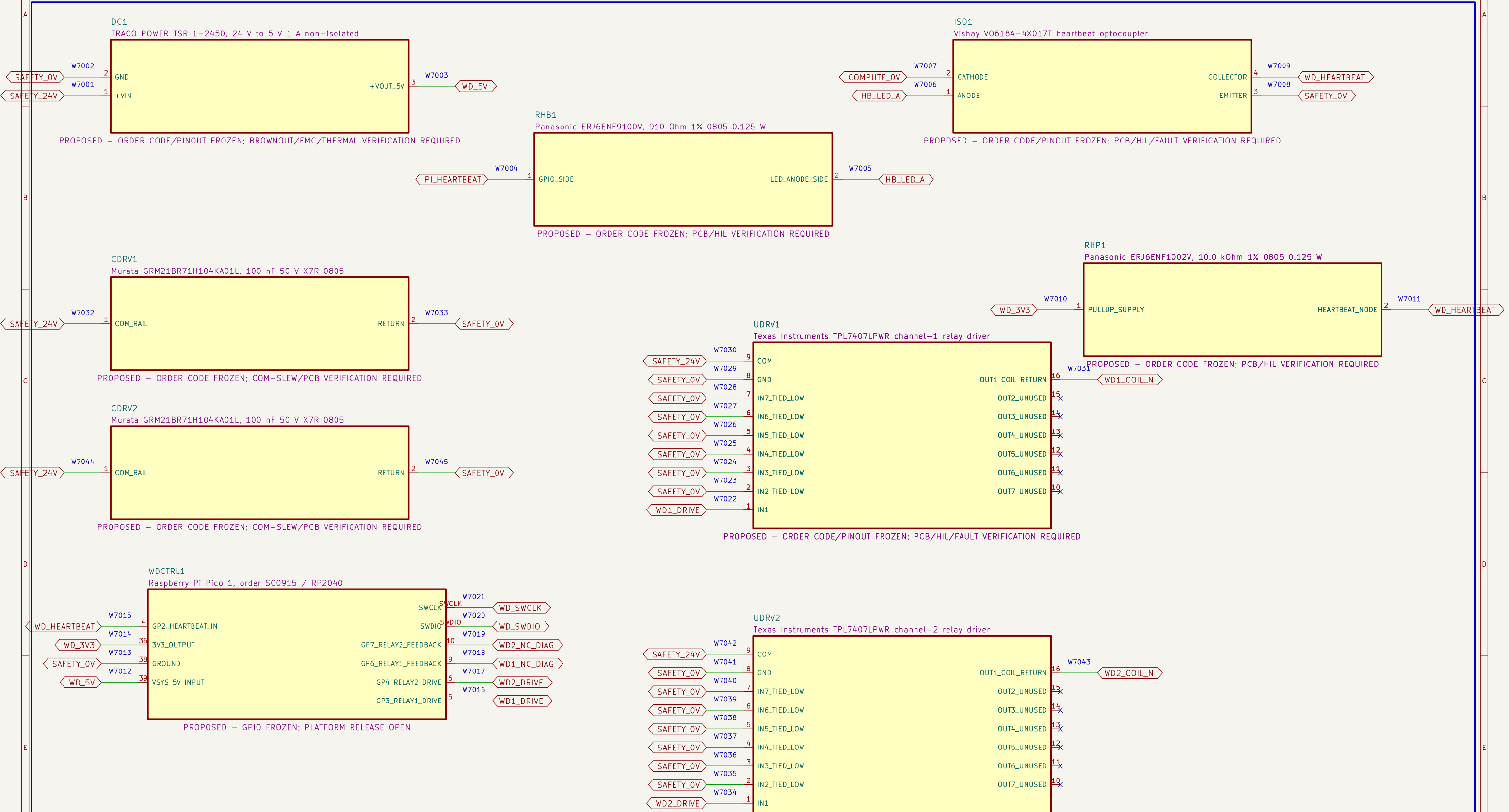
NOTE 1: Standard fully populated ROBOTIS TTL daisy-chain cables are prohibited because they would parallel branch VDD.

NOTE 2: Every injection module requires released source, continuity, isolation, pull and no-backfeed tests.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /06 Protected actuator branches and VDD-isolating injection/
File: 06_branches_and_injection.kicad_sch

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



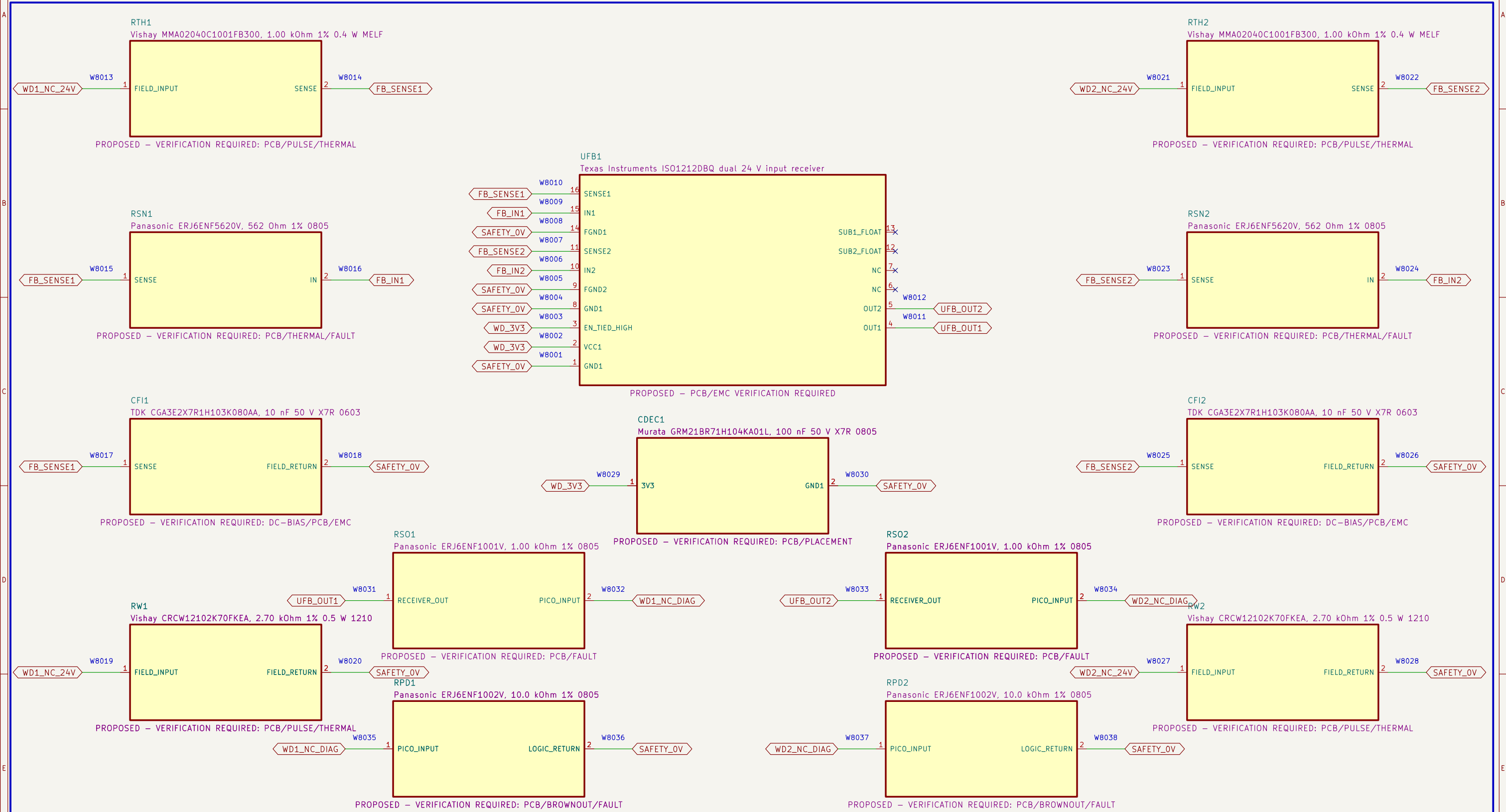
NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only:
current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /07 Independent watchdog power, controller and drivers/ File: 07_watchdog_control.kicad_sch		
Title: PB HR–V0 Electrical V3 – 07		
Size: A3	Date: 2026–08–06	Rev: V3–P0.9
KiCad E.D.A. 10.0.5		Id: 8/11

08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.

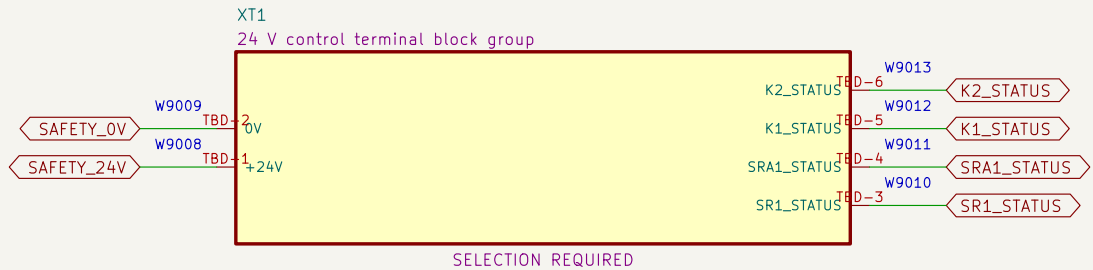
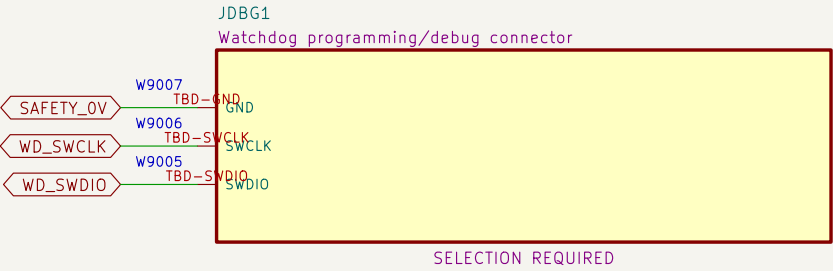
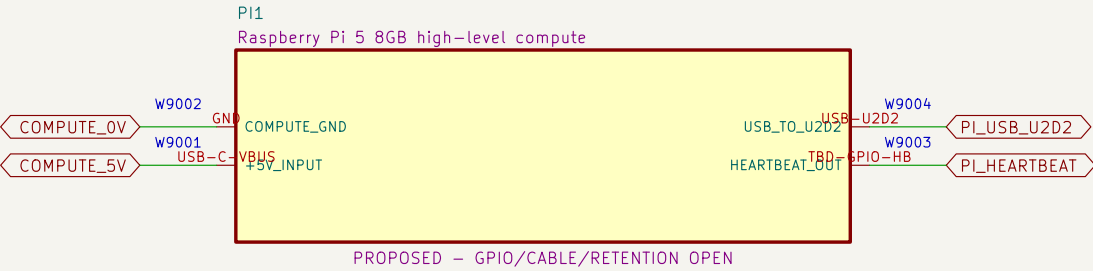


NOTE 1: Type-3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is 2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_OV; the ISO1212 barrier is not credited. Passive identities are frozen; PCB, derating, EMC and fault tests remain open.

09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.

